Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 38

Title: Metabolic changes in Diabetes mellitus

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Objectives: Metabolic changes in Diabetes mellitus

SOM. MK.I.BPM2.3.DM.3.BCHM.1361	Differentiate between type 1 and type 2 diabetes mellitus based on clinical presentation and laboratory findings.
SOM. MK.I.BPM2.3.DM.3.BCHM.1362	Correlate presenting features and metabolic basis for clinical features in patients with diabetes mellitus
SOM. MK.I.BPM2.3.DM.3.BCHM.1363	Explain metabolic basis for hyperglycemia while reviewing pathways and enzymes of carbohydrate metabolism that are active in liver in an uncontrolled diabetic.
SOM. MK.I.BPM2.3.DM.3.BCHM.1364	Explain metabolic changes in diabetic ketoacidosis (type 1 diabetics). Highlight occurrence of ketosis, metabolic acidosis (high anion gap) and hyperglycemia. Indicate changes in serum potassium following insulin therapy.
SOM. MK.I.BPM2.3.DM.3.BCHM.1365	Name the major ketone body found in blood and urine in diabetic ketoacidosis. Review enzymes of lipolysis, beta-oxidation and ketogenesis that are active in uncontrolled type 1 diabetic.
SOM. MK.I.BPM2.3.DM.3.BCHM.1366	Analyze metabolic basis for dehydration, hyperglycemia and hyperosmolarity in hyperosmolar hyperglycemic state in type 2 diabetics.
SOM. MK.I.BPM2.3.DM.3.BCHM.1367	Compare and contrast biochemical and laboratory findings in hyperosmolar hyperglycemic state and diabetic ketoacidosis
SOM. MK.I.BPM2.3.DM.3.BCHM.1368	Compare and contrast microvascular and macrovascular complications in diabetes mellitus
SOM. MK.I.BPM2.3.DM.3.BCHM.1369	Explain biochemical basis for mechanism of microvascular changes (sorbitol, AGE formation, role of DAG, hexosamine pathway) for nephropathy, neuropathy and retinopathy, Peripheral arterial disease and diabetic foot changes
SOM. MK.I.BPM2.3.DM.3.BCHM.1370	Identify changes in serum lipids (dyslipidemia) responsible for macrovascular complications
SOM. MK.I.BPM2.3.DM.3.BCHM.1371	Evaluate laboratory tests for management of diabetes mellitus. Analyze significance of microalbuminuria in identification of early nephropathy

Recommended Reading

- Lippincott's Biochemistry 9th edition; Chapter 25: Diabetes mellitus
🔗 https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402
- Q in online Lippincott's Biochemistry (Q on hypoglycemia and Diabetes mellitus)
- Study Questions: 25.1 – 25.4
- Concept map Pg: 347
- Review Lecture ER – Diabetes and Hypoglycemia
- **Slides 6-15 are review slides**

Metabolic changes in Diabetes mellitus

- Definition
- Types
- Presenting features
- Complications
  - Acute
  - Chronic
- Laboratory tests for diagnosis and long-term management

Types of Diabetes

- Includes a heterogenous group of **multifactorial** disorders characterized by the presence of **hyperglycemia** (high blood glucose levels) that result from defects in secretion of insulin OR action of insulin OR both
- Type 1 diabetes (Insulin dependent diabetes mellitus)
- Type 2 diabetes (Non-insulin dependent diabetes mellitus)
- Gestational diabetes
  - Diabetes during pregnancy and blood glucose levels normalize following delivery
  - Higher incidence of fetal malformations, macrosomia and neonatal hypoglycemia
- Secondary diabetes:
  - Secondary to other disorders like excessive secretion of cortisol, growth hormone, chronic pancreatic disorders

Type 1 Diabetes mellitus

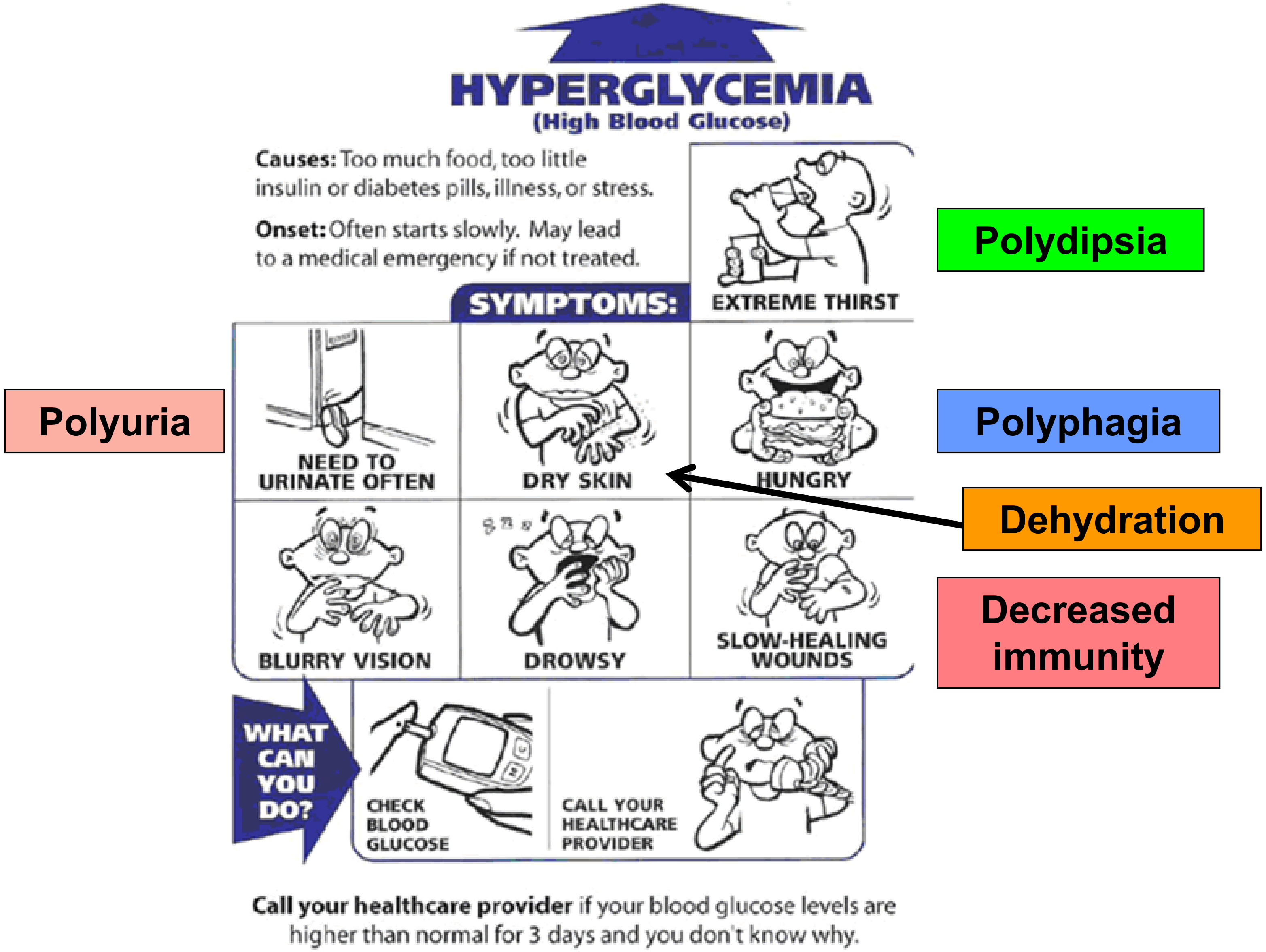
- Insulin dependent diabetes mellitus (IDDM)/ Juvenile onset diabetes mellitus
- **Autoimmune destruction** of the β-cells of islets of pancreas, resulting in destruction of the islet cells and a **marked reduction in insulin secretion**
- Insulin levels are very low or absent
- Represents about 10% of all patients with diabetes mellitus
- Presents during adolescence (young adults)
- Patients have to be on life-long insulin supplementation to prevent the complications of diabetes

Type 2 Diabetes mellitus

- Non-insulin dependent diabetes mellitus (NIDDM)
- **Obesity (Syndrome X/ Metabolic syndrome/ insulin resistance syndrome)** is an important risk factor for development of type 2 diabetes mellitus and insulin resistance
- Target tissues for insulin (liver, skeletal muscle & adipose tissue) do not respond to circulating insulin **(insulin resistance)** and there is a decrease in insulin secretion with time (**beta cell dysfunction**)
- Circulating insulin levels are high/ normal/ low (depending on the stage of the disease)
- Usually respond to oral hypoglycemic agents and insulin may be required in the later stages of the disorder
- Type 2 diabetic patients have
  - Insulin resistance

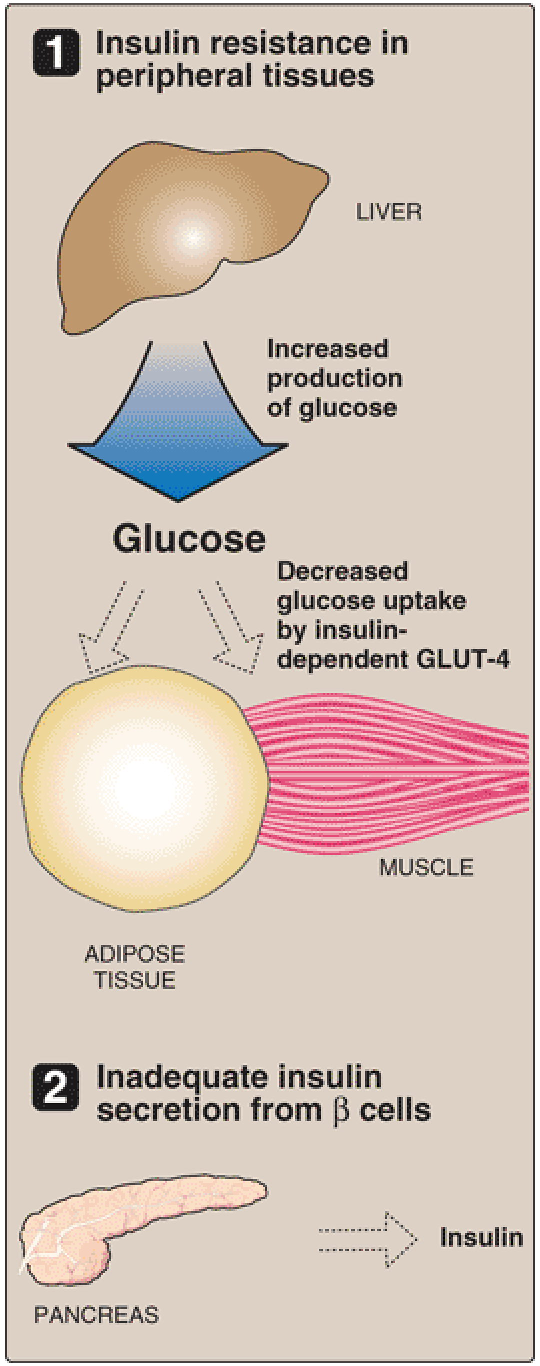
Beta cell dysfunction and reduced secretion of insulin

Presenting features of Diabetes mellitus



Presenting features

- Classical triad: Polyphagia, Polydipsia, and Polyuria are more common with type 1 diabetes
- Weight loss may be observed in many patients (type 1) – due to accelerated lipolysis and muscle proteolysis
- Many of patients with type 2 diabetes are obese (insulin resistance)
- Diabetes mellitus affects carbohydrate, lipid and protein metabolism
- Decreased secretion of insulin/ ineffective insulin action → Hyperglycemia



Mechanism of hyperglycemia in diabetes mellitus (Type 1 and type 2)

Increased gluconeogenesis in liver

Decreased number of GLUT4 in peripheral tissues

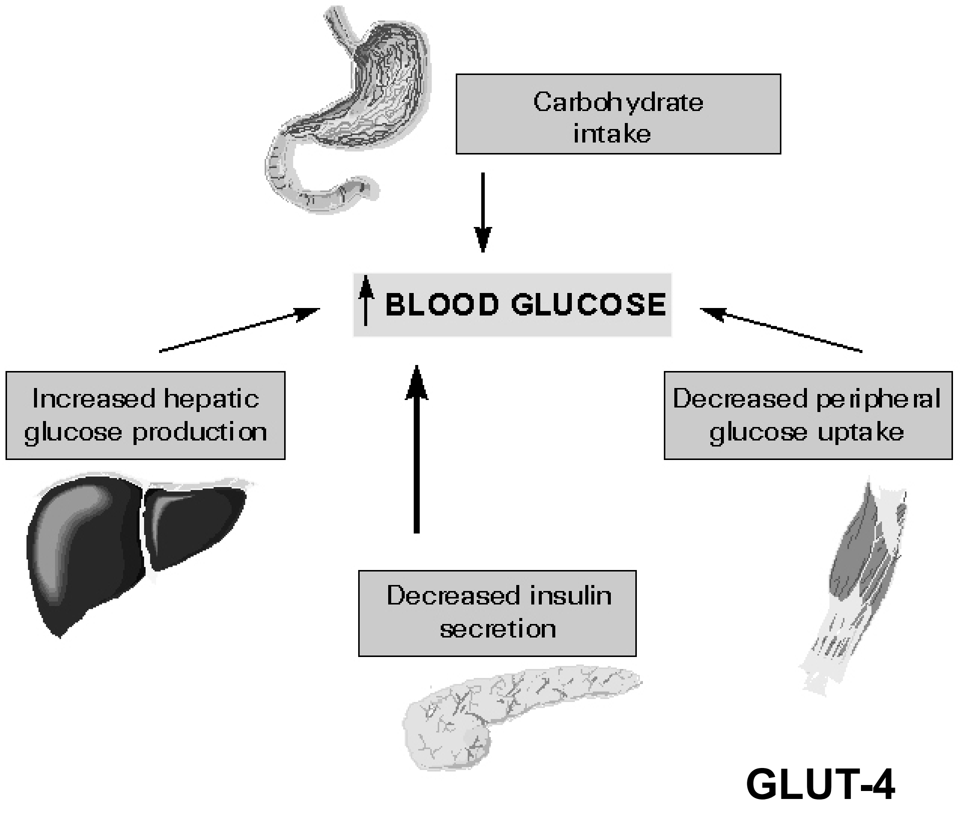
Decreased number of insulin receptors/ post receptor defects

Decreased secretion of insulin from the pancreas

Why is there hyperglycemia???

http://emedicine.medscape.com/article/117853-overview#a0104 Type 2 diabetes mellitus
http://emedicine.medscape.com/article/117739-overview Type 1 diabetes mellitus

- In an uncontrolled diabetic the **insulin/ glucagon ratio** is low
- Key **gluconeogenic** enzymes are activated (induced)
  - Phosphoenolpyruvate carboxykinase (PEPCK)
  - Pyruvate carboxylase
  - Fructose 1,6- bisphosphatase (low levels of fructose 2,6-bisphosphate)
  - Glucose 6-phosphatase
  - **Amino acids** from muscle proteolysis are used for gluconeogenesis
- Glycolysis is inhibited (liver)
- Glycogenesis is inactive **(liver and muscle)**



Insulin dependent uptake of glucose in adipose tissue and muscle (GLUT4)

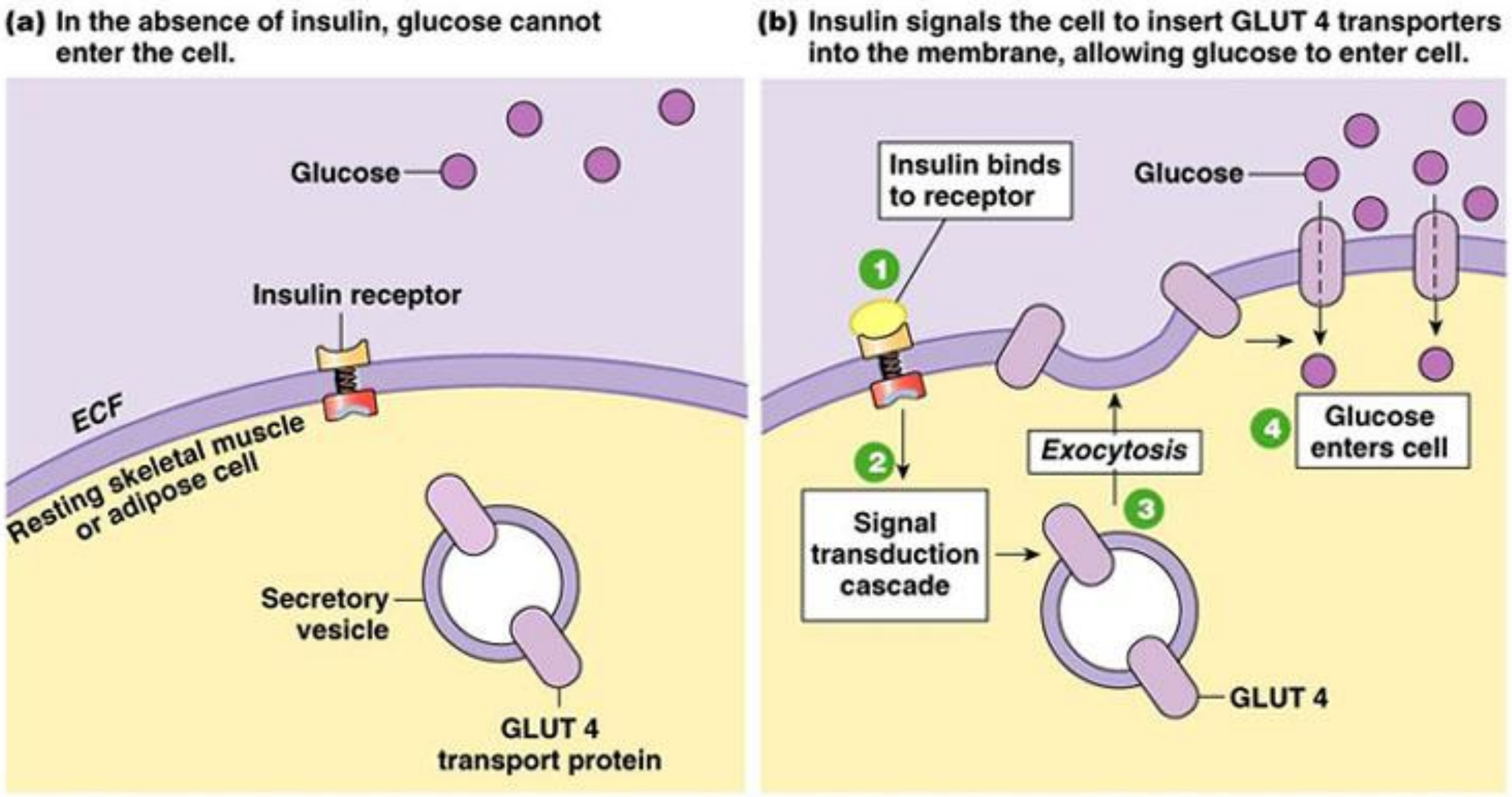
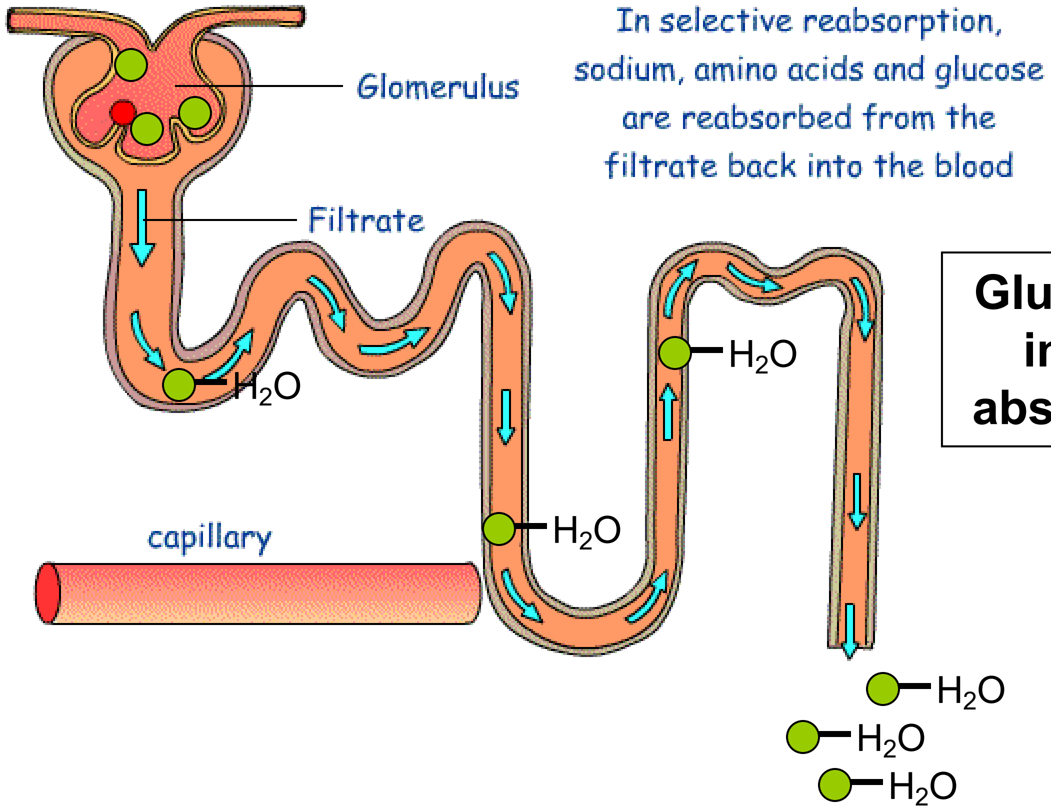


Fig. 22-12

The deficiency of insulin/ insulin resistance results in decreased number of **GLUT4** in peripheral tissues (muscle and adipose tissue), contributing to hyperglycemia.

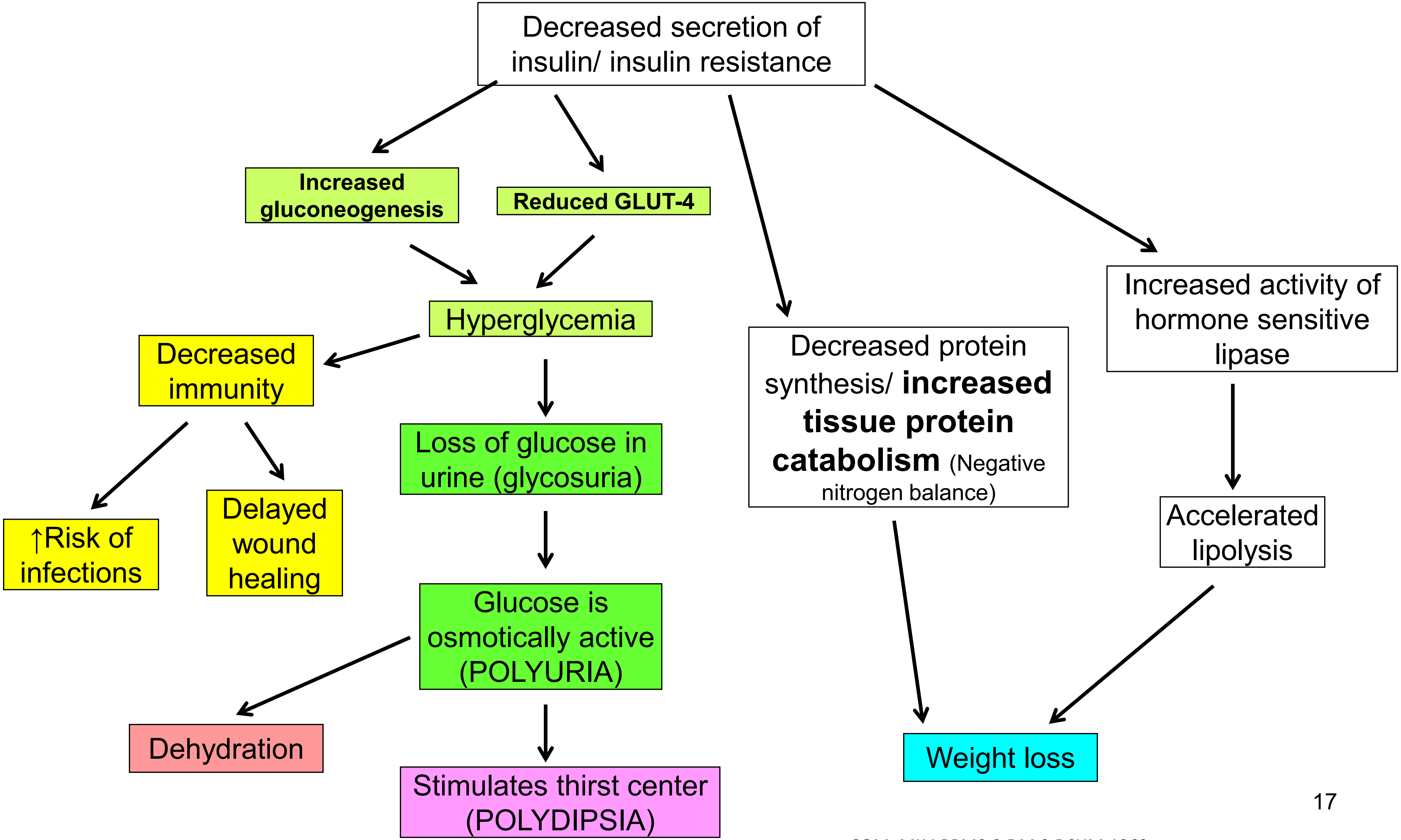
Exercise results in an insulin independent uptake of glucose by muscle via **AMP-kinase pathway**

Basis of glycosuria and 'polyuria'



Glucose is completely reabsorbed in the renal tubule. Glucose is absent in urine in a normal person

In hyperglycemia, a large amount of glucose is filtered, that exceeds the reabsorptive capacity of the tubule. Glucose is osmotically active (holds water), resulting in 'polyuria' (Osmotic diuresis)


Decreased secretion of insulin/ insulin resistance
Increased gluconeogenesis
Reduced GLUT-4
Increased activity of hormone sensitive lipase
Hyperglycemia
Decreased immunity
Decreased protein synthesis/ **increased tissue protein catabolism** (Negative nitrogen balance)
Loss of glucose in urine (glycosuria)
↑Risk of infections
Delayed wound healing
Accelerated lipolysis
Glucose is osmotically active (POLYURIA)
Dehydration
Weight loss
Stimulates thirst center (POLYDIPSIA)

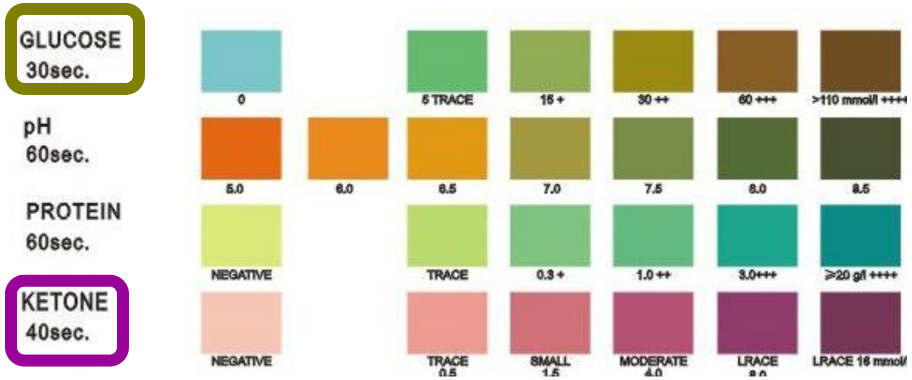
Complications of diabetes mellitus

- **ACUTE**
  - Ketoacidosis (type 1)
  - Hyperosmolar hyperglycemic state (type 2)
  - Hypoglycemia in diabetics on treatment
- **CHRONIC**
  - Microvascular
  - Macrovascular

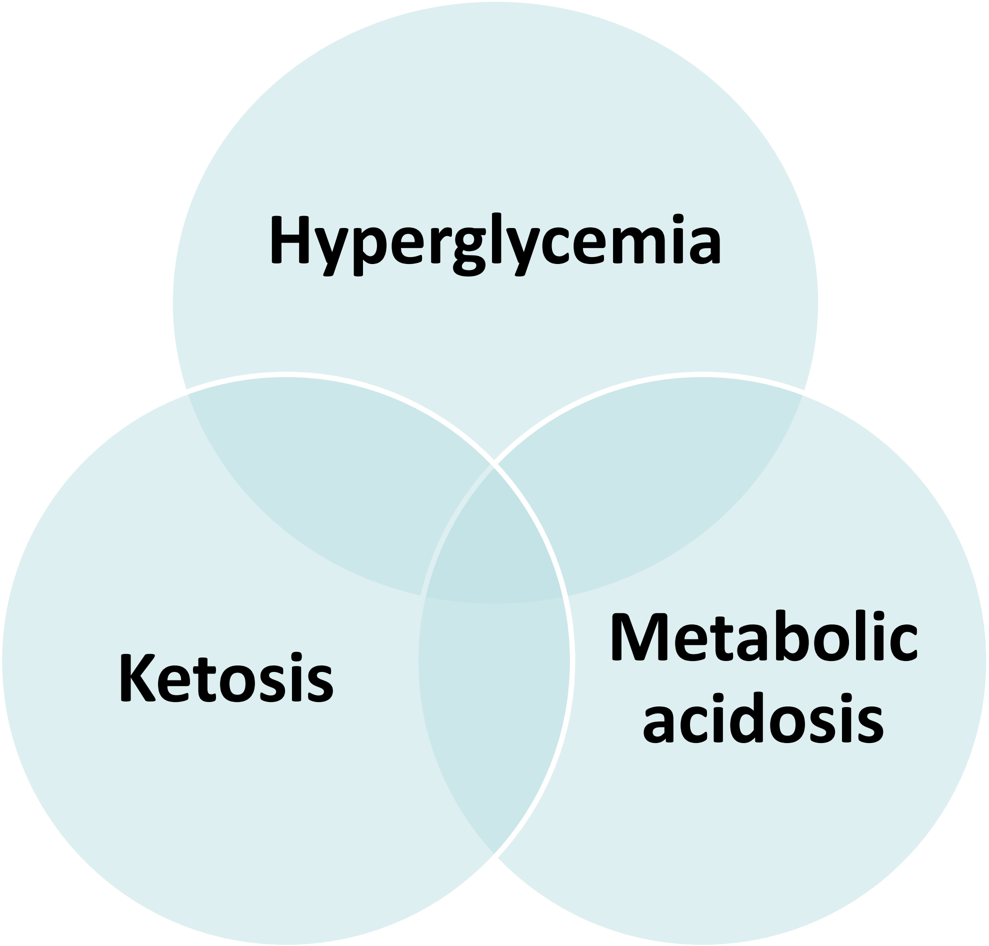
Diabetic ketoacidosis

- An 18-year-old girl complained of tiredness and weight loss and feeling drowsy. She admitted to feeling thirsty and passing large quantities of urine. She would get up at night to pass urine.
- She was found to have BP 95/65, pulse 110/min and cold extremities. She had deep sighing respiration (Kussmaul breathing) and her breath smelt of acetone 'nail varnish/ fruity odor'
- Lab tests:
  - Plasma glucose = 340 mg/dL
  - Arterial pH: 7.1; Serum HCO_3: 15 mEq/L; PCO_2: 25 mm Hg
  - Increased anion gap (High levels of 3-hydroxybutyrate)
  - Urine dipstick test revealed glycosuria and ketonuria.

Features of diabetic ketoacidosis



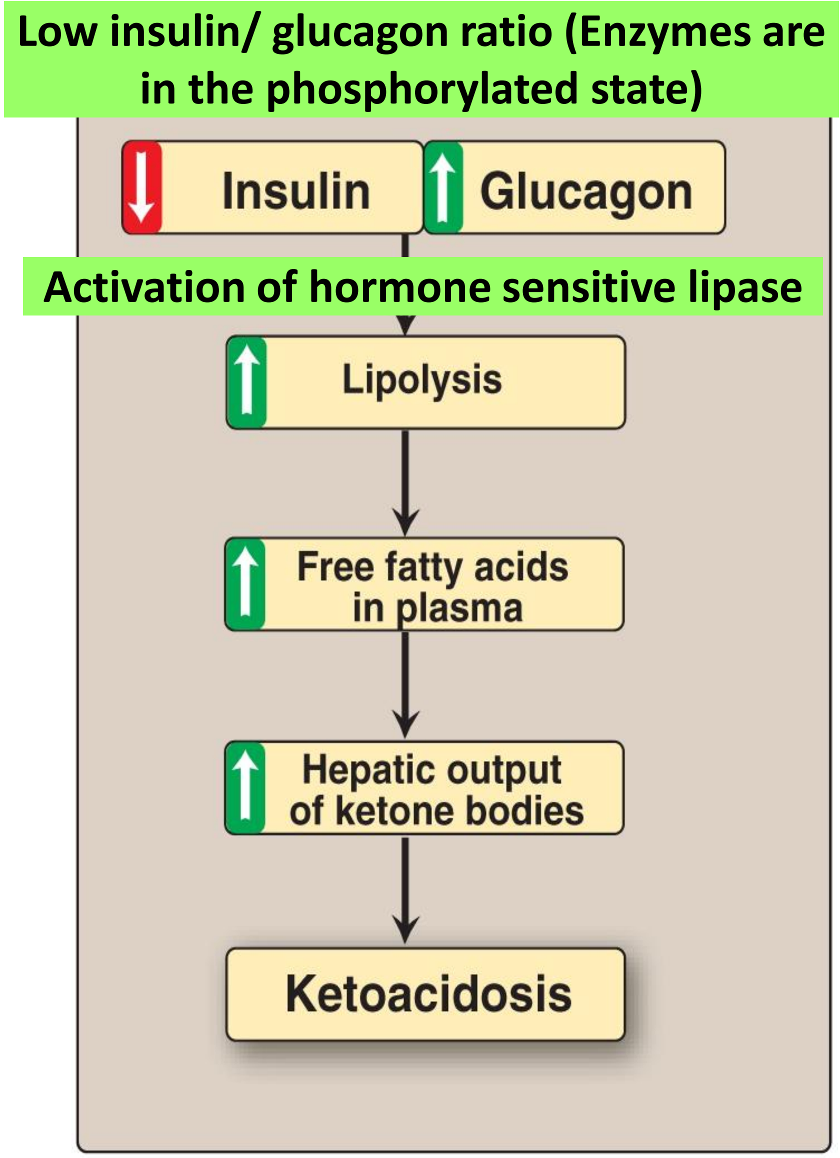
Urine dipstick tests



Ketosis in type 1 diabetes mellitus

- Insulin deficiency results in uncontrolled adipose tissue lipolysis (Hormone sensitive lipase is active due to low insulin/glucagon ratio; HS lipase is in the phosphorylated form)
- There is increased free fatty acid delivery to the liver
- Increased rates of β-oxidation (Active CPT-1 and increased free fatty acids), results in the formation of acetyl CoA, that is used for ketone body synthesis (active **mitochondrial HMG CoA synthase** and HMG CoA lyase)
- Ketogenesis in liver >> peripheral utilization of ketone bodies

The ketone body formed in highest concentration is **3-hydroxybutyrate (major ketone body)**



Metabolic acidosis

- Excessive ketone bodies are excreted in urine (ketonuria) – can be detected by dipsticks. **3-hydroxybutyrate** and acetoacetate in blood and urine
- Increased formation of **3-hydroxybutyrate** and acetoacetate; Ketone bodies are weak acids; Bicarbonate levels fall as they are used for buffering the excessive protons produced (Metabolic acidosis)
- **Increased anion gap** (due to very high levels of ketone bodies)
- pH: Low; HCO_3^-: Low; PCO_2: Low
- Compensation by the respiratory system→ ↑rate and depth of ventilation (hyperventilation) – Kussmaul breathing
- Kidneys can also compensate by excreting an acidic urine after 3-5 days (increased ammonium and phosphate excretion in urine)
- Ketone bodies are also lost via lungs (acetone) – 'fruity odor of breath'
- The metabolic acidosis can be severe and may worsen coma (Ketoacidotic coma)

Mechanism of **Diabetic ketoacidosis**

Destruction of β-cells of pancreas

↓

Very low secretion of insulin

- Increased muscle proteolysis → Amino acids → Increased liver gluconeogenesis → Hyperglycemia
- Increased liver gluconeogenesis → Hyperglycemia
- Reduced GLUT-4 → Hyperglycemia
- Increased hormone sensitive lipase activity → Increased adipose tissue lipolysis → Increased ketogenesis in liver → Ketosis → Metabolic acidosis (High anion gap)

Hyperglycemia

↓

Loss of glucose in urine (glycosuria)

↓

Glucose is osmotically active (POLYURIA)

- → Dehydration
- → Stimulates thirst center (POLYDIPSIA)

Ketosis

↓

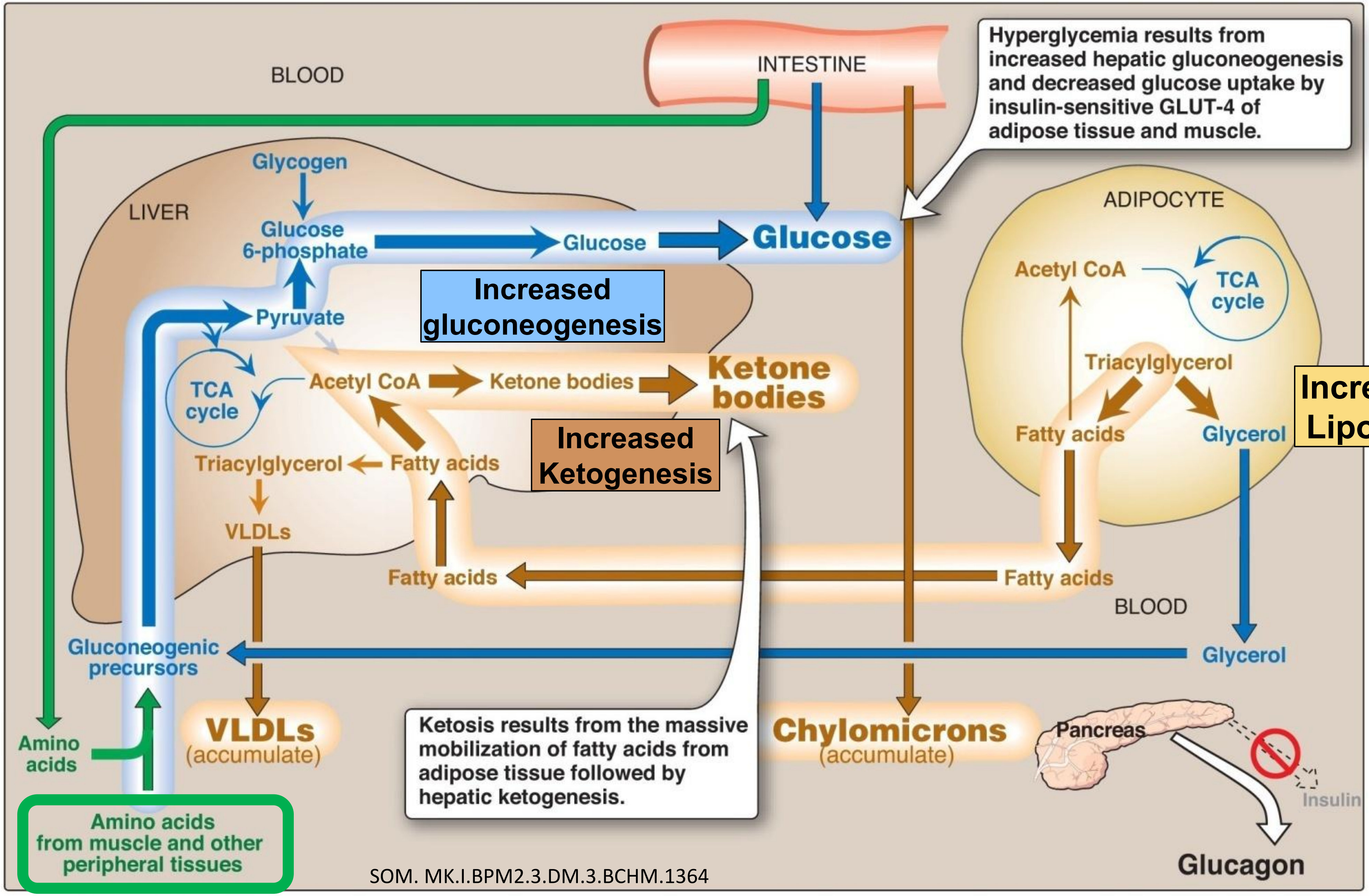
Metabolic acidosis
(High anion gap)

Effects of hyperglycemia

- Elevated blood glucose levels → Glycosuria results in increased water loss → Dehydration → Stimulation of the thirst center (polydipsia)
- Coma is usually due to hyperglycemia (Osmotic effect: Water moves out of ICF resulting in neuronal dehydration) and is worsened by metabolic acidosis
- Infection/ illness exacerbates the predisposition to ketoacidosis (insulin requirements increase during an episode of illness due to insulin resistance due to stress)
- Failure to comply with insulin therapy may also result in ketoacidosis in type 1 diabetics

Ketoacidosis may be the presenting feature in many patients with type 1 diabetes

Metabolic changes in type 1 diabetes mellitus



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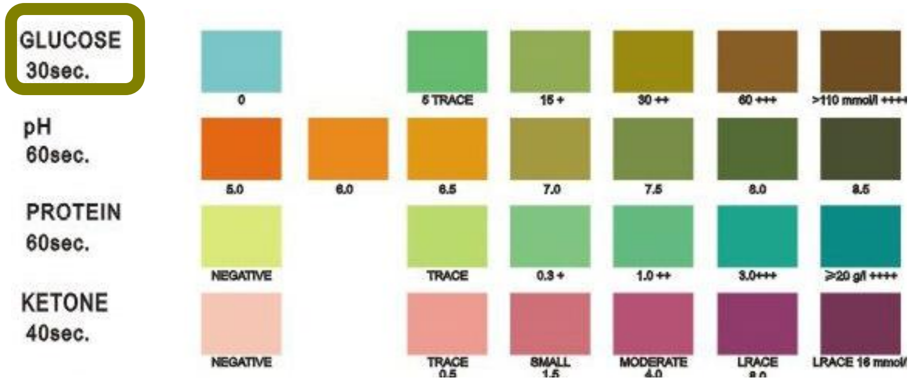
Fluid and electrolyte changes

- **Dehydration** – Due to loss of water due to osmotic diuresis due to glucosuria (glycosuria)
- Changes in serum **potassium**: (Hyperkalemia) Insulin deficiency and acidosis result in a shift of potassium from the ICF to the ECF. This results in loss of potassium in urine (Whole body potassium is reduced, even though serum potassium may be high).
- When insulin is injected during the treatment, potassium moves back into the ICF – Since potassium reserves are low, patient is in danger of **hypokalemia** (requires close monitoring)

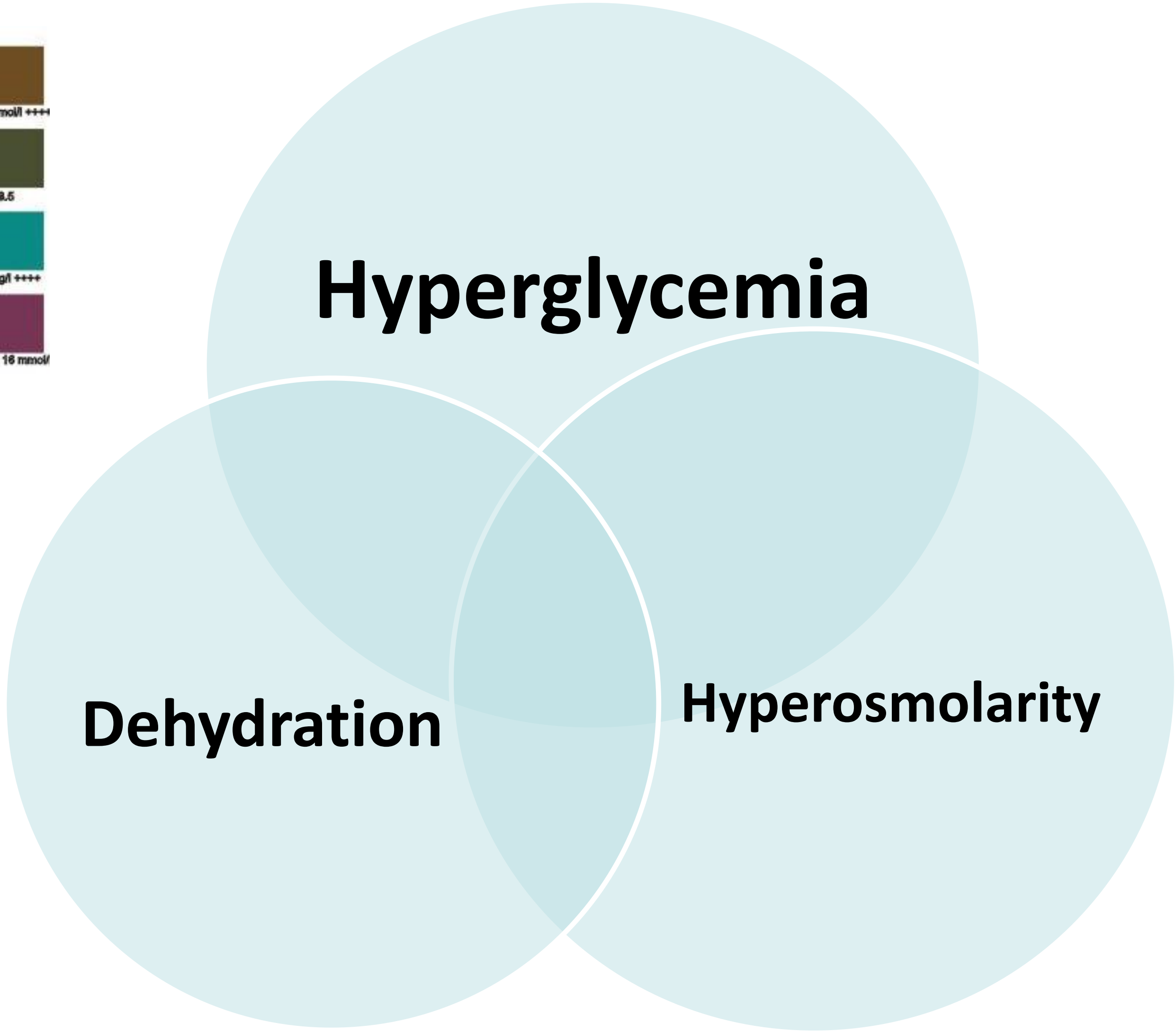
Hyperosmolar Hyperglycemic state

- A 66-year-old woman, known type-2 diabetic was brought to the emergency department because of fever, chills, arthromyalgia, and fatigue.
- At presentation, she was confused, lethargic, and **dehydrated**.
- Initial laboratory studies revealed
  - **Plasma glucose**: 916 mg/dl (<125)
  - creatinine 1.7 mg/dl
  - sodium 146 mEq/l
  - potassium 5.3 mEq/L (3 -5)
  - Arterial pH: 7.4
  - **Plasma osmolarity:** 342 mOsm/l (N: 280-300)
  - Urinalysis was negative for ketones but positive for glucose

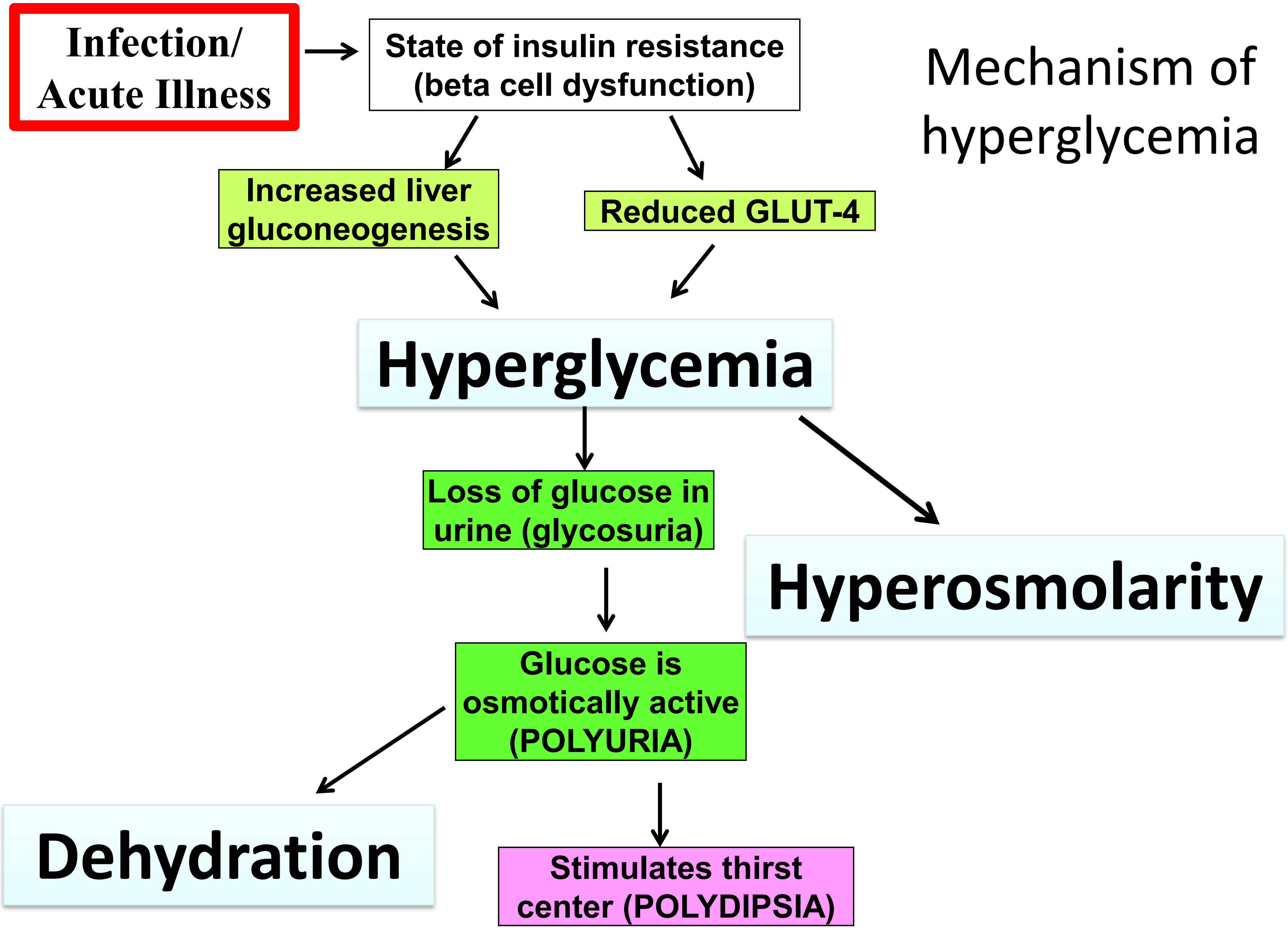
Features of hyperosmolar hyperglycemic state



Urine dipstick tests



Mechanism of hyperglycemia



Hyperosmolar hyperglycemic state

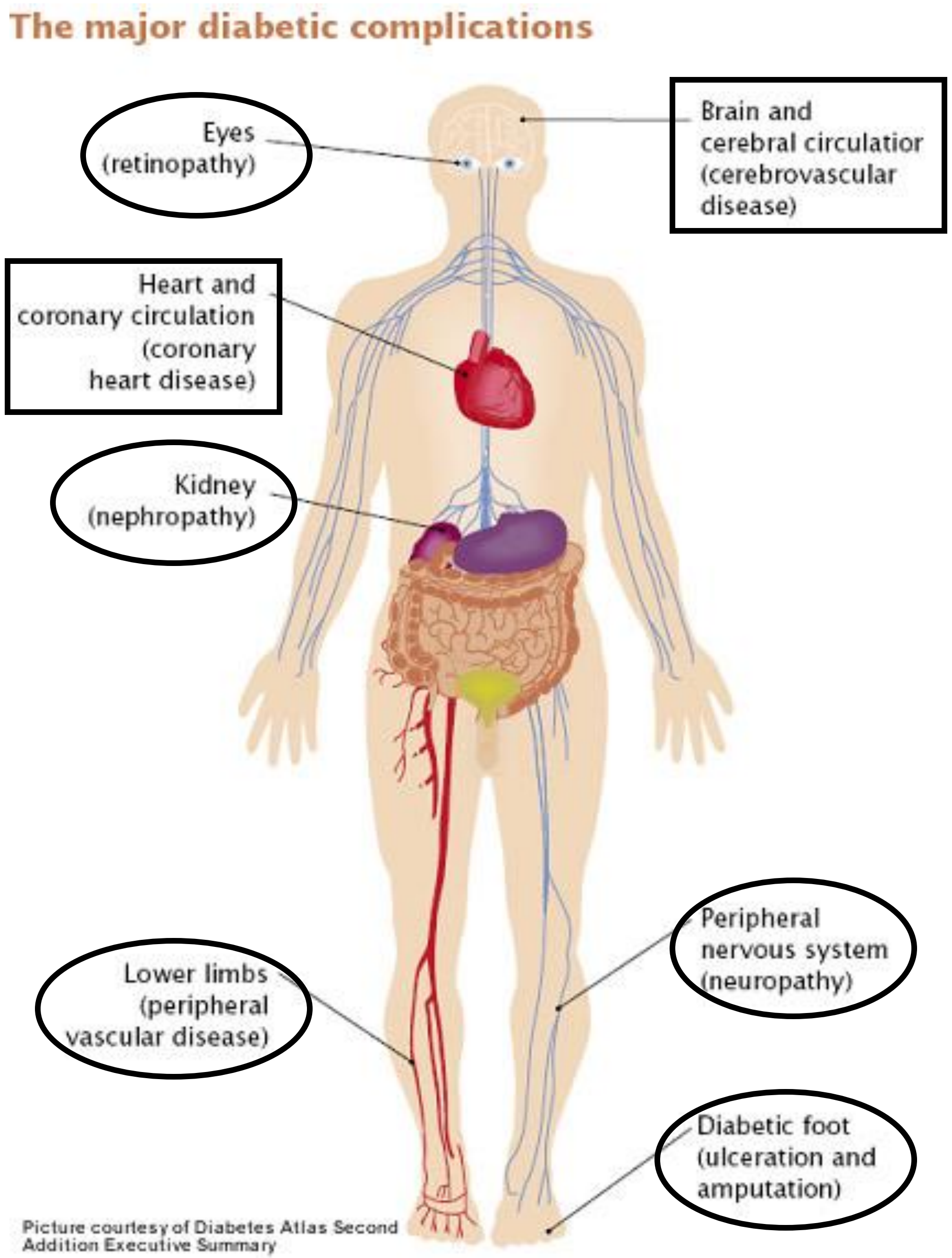
- More common in Type 2 diabetes mellitus
- Typically precipitated by an acute illness/ infection (insulin resistance)
- Hyperglycemia: Plasma glucose levels are markedly elevated (>600mg/dL)
- Hyperosmolarity: Osmolarity of blood is **increased** due to the high blood glucose levels (>320 mOsm/L; Normal-280-300)
- There is polyuria (glycosuria) – excessive losses of water in urine – osmotic diuresis (Polyuria)
- Patients have a low blood volume as a result of polyuria (severe dehydration)
- Ketone body production is NOT significant (compare to diabetic ketoacidosis).
- Why is ketosis NOT a prominent feature of type 2 diabetes mellitus??? – Patients have some insulin secretion which inhibits ketogenesis
- Metabolic acidosis NOT commonly associated

Hyperosmolar hyperglycemic state

- Intracellular dehydration in the neurons, as ICF water moves into ECF (High plasma osmolarity; Glucose is osmotically active) → Changes in hydration of neurons → Neurological deficits, unconsciousness (coma)
- Neurological changes usually revert back to normal on management of hyperglycemia
- Fluid balance to be restored during management to correct the dehydration

Chronic complications

- Chronic complications are **common to type-1 and type-2 diabetes mellitus**
- The incidence of chronic complications is higher in patients who have higher plasma glucose levels and higher HbA1c levels (Poor glycemic control)
- **Microvascular**: Eyes, retina, neurons, kidney
- **Macrovascular:** Atherosclerosis and its complications due to dyslipidemia



Chronic complications

Microvascular

Macrovascular

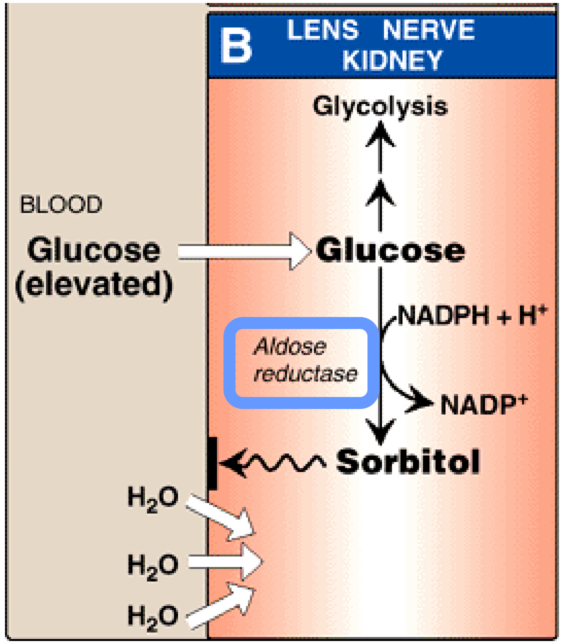
Microvascular complications

- Neuropathy, nephropathy and retinopathy are the microvascular complications of diabetes
- Occurs in tissues which do not require insulin for glucose entry (for example retina, nervous tissue, lens, nephrons)
- Mechanisms of complications: Related to high intracellular glucose and/or its metabolites
  - Sorbitol formation
  - Nonenzymatic glycation of proteins (Advanced glycation end products: AGE)
  - Hexosamine pathway
  - DAG and protein kinase C
- All the processes are dependent on the plasma glucose concentration

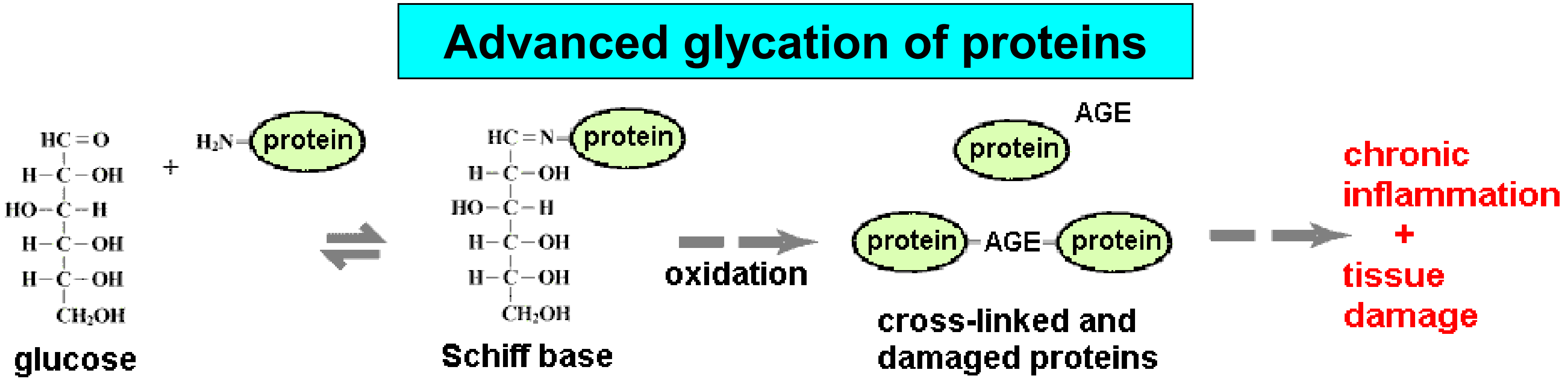
Microvascular complications: **Sorbitol**

- In the lens, nerve and kidney, sorbitol formed is osmotically active and results in cell swelling due to water retention
- The formation of sorbitol is directly related to the plasma glucose levels
- Enzyme: Aldose reductase
- Increased sorbitol formation in the **lens** results in increased water content of the lens that can result in vision changes and cataracts (due to changes in solubility of proteins).
- Sorbitol accumulation in neurons affects function of nerves

Polyol pathway



Microvascular complications: **Advanced glycation end products (AGE)**

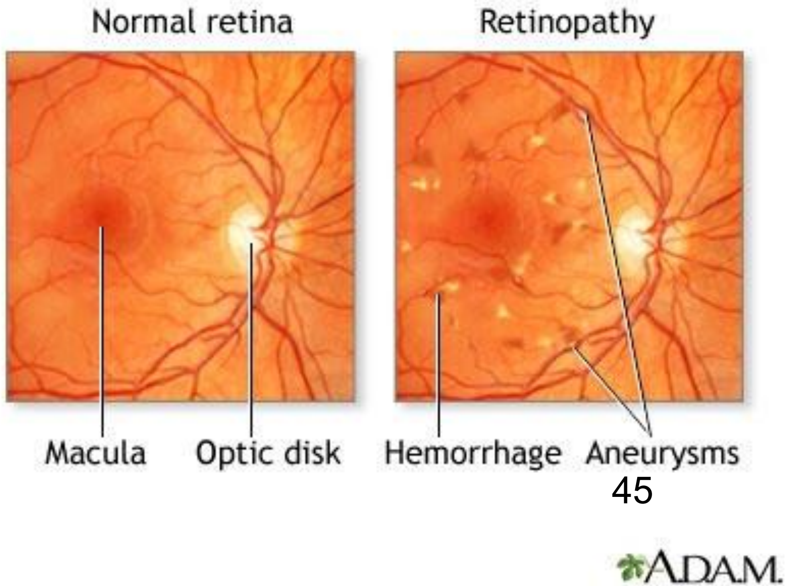


- Non-enzymatic addition of glucose to intracellular and extracellular proteins (Process similar to formation of HbA1c)
- Glycation of proteins results in cross-linking of glycated proteins and altered function – leading to **glomerular dysfunction, endothelial damage and changes in extracellular matrix**
- AGE modified proteins are more prone for oxidative damage

Microvascular complications: **Advanced glycation end products (AGE)**

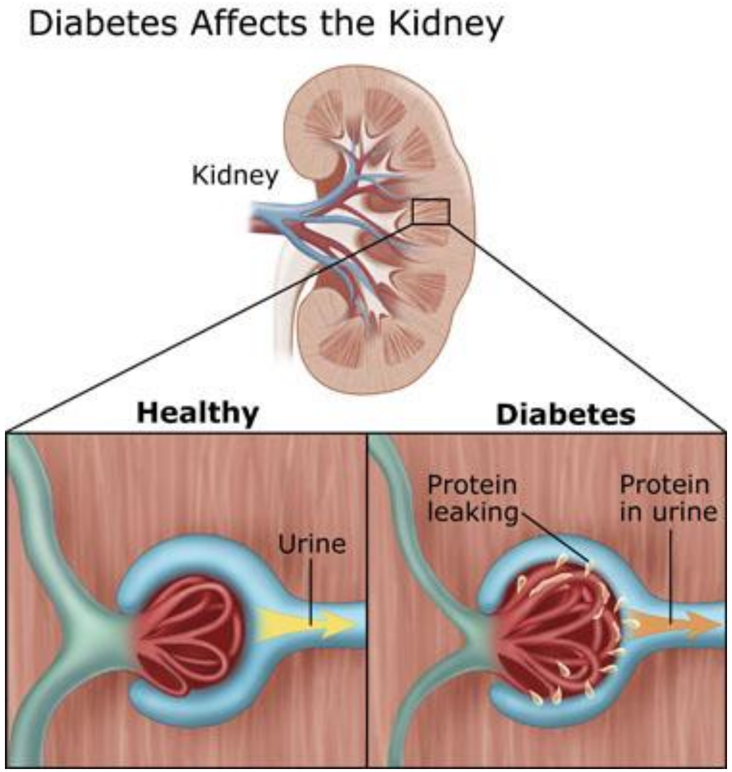
- AGEs (advanced glycation end products) alter metabolism and function of **neuronal** proteins
- AGEs bind to their receptors (RAGEs) and cause release of proinflammatory molecules
- Alteration of the glomerular basement membrane proteins of **nephron** – loss of albumin in urine (detected as **microalbuminuria**)

Diabetic retinopathy



Microvascular complications: AGEs

Foot ulcers: **Painless** (neuropathy and peripheral vascular disease are contributory factors)



Increased loss of albumin in urine in the initial stages (**microalbuminuria**); Earliest indicator of diabetic renal involvement

In later stages, there may be development of renal failure

Microvascular complications: **DAG pathway**

- Hyperglycemia increases the formation of DAG (Diacylglycerol) which activates protein kinase C
- Activation of protein kinase C increases transcription of proteins like fibronectin, collagen and proteins of the extracellular matrix
- This results in changes in the neurons, nephrons and endothelium of blood vessels

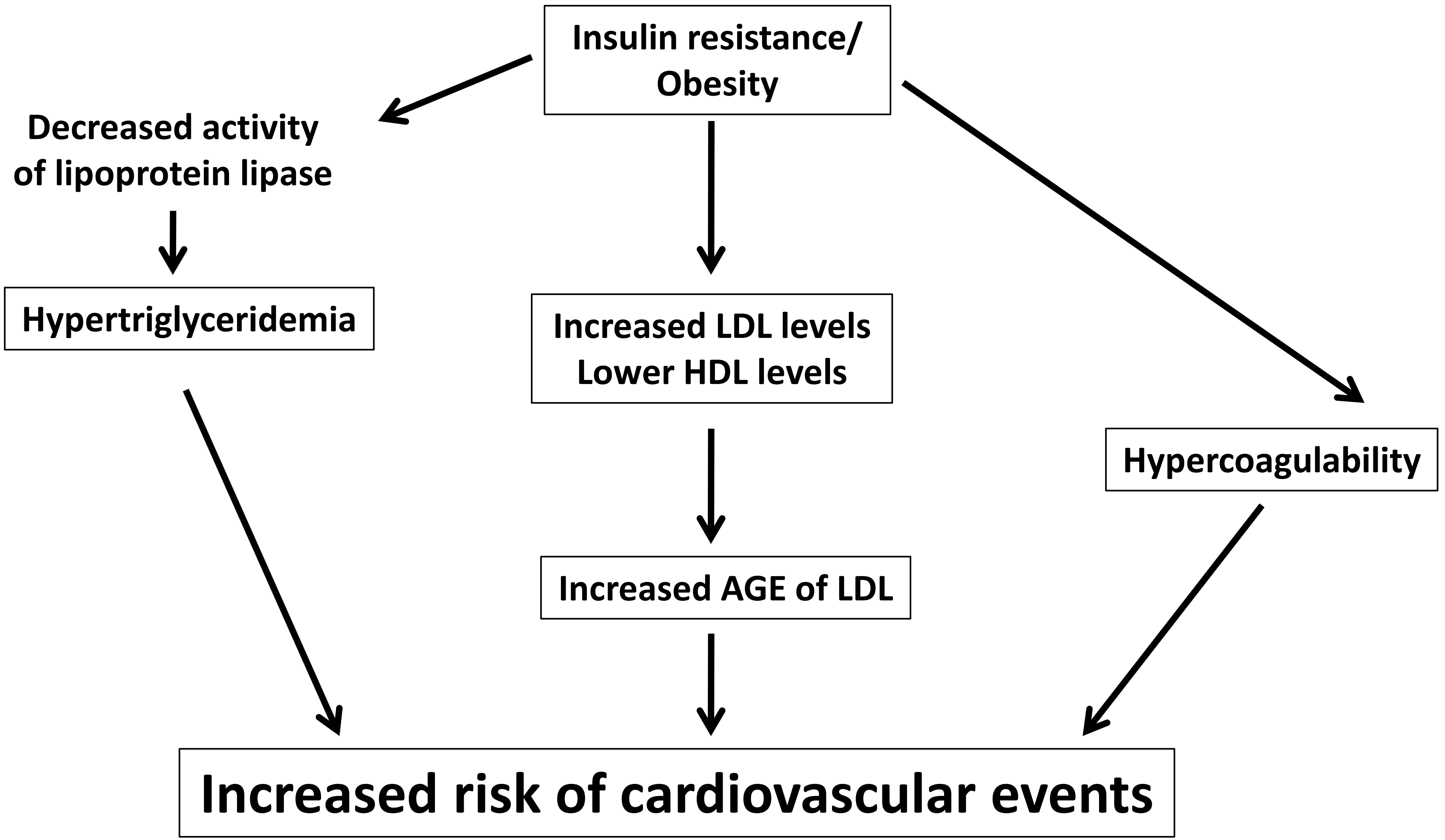
Microvascular complications: **Hexosamine pathway**

- Hyperglycemia increases flux through the hexosamine pathway and forms fructose 6-phophate and hexosamines (glucosamine; aminosugars)
- Hexosamines are substrates for proteoglycan synthesis and glycoprotein synthesis
- This results in altered function of these components and may cause altered expression of certain growth factors (?)

Macrovascular complications due to dyslipidemia

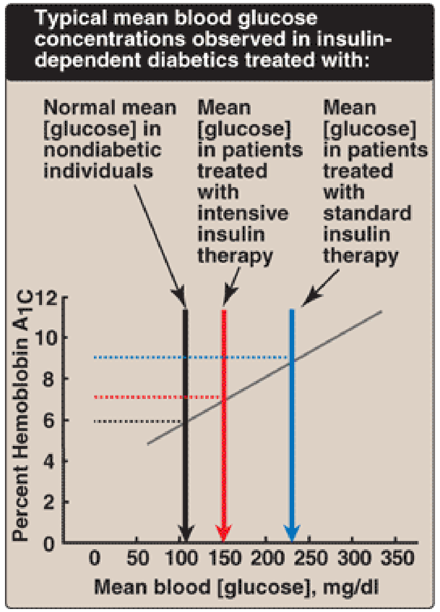
- Accelerated atherosclerotic changes resulting in cardiovascular disease or stroke (Macrovascular complications)
- Patients with both types of diabetes have **hypertriglyceridemia** (↑ circulating triglycerides). There is increased chylomicrons and VLDL in the circulation.
- Due to the decreased action of endothelial **lipoprotein lipase** (lipoprotein lipase is induced by insulin)
- Obese subjects with diabetes mellitus also may have hypertension, dyslipidemia and hypercoagulability (insulin resistance/ syndrome X) – all these factors increase the risk of cardiovascular disease in diabetes
- Atherogenic small dense **LDL levels** are higher in type-II diabetics
- HDL levels are lower in type-2 diabetics
- **AGEs** are also implicated in atherogenesis (glycation of LDL and glycation of proteins in the blood vessel wall)

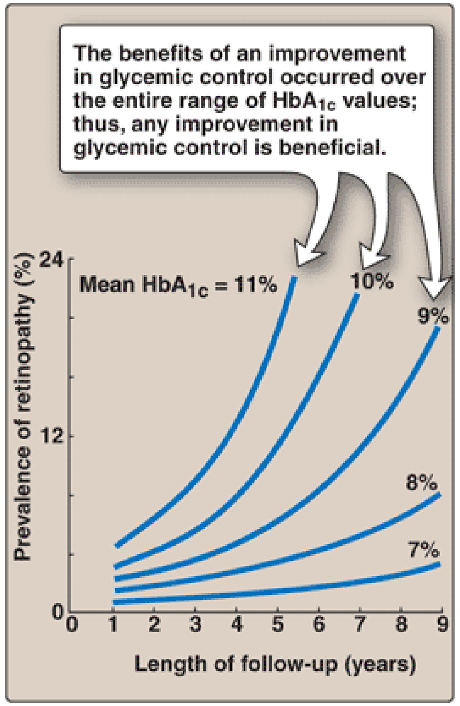
Dyslipidemia in type-2 diabetes mellitus



HbA1C and its significance

- HbA1c is the non-enzymatic glycation of hemoglobin (HbA1c levels depends on the blood glucose concentration)
- HbA1c levels are an indicator of long term glucose levels (Over the past 3-4 months)
- Patients that have poor blood glucose control (reflected by high HBA1C levels), have been observed to have a higher incidence of complications (microvascular and macrovascular)
- Optimal blood glucose control reduces the incidence of complications
- HbA1c is increasingly also used for the diagnosis of diabetes mellitus
- Levels greater than 6.5% are indicative of diabetes mellitus





Laboratory tests for long term management

Tests for glycemic control

Fasting blood glucose

Postprandial blood glucose

Glycated hemoglobin levels

Tests for lipid profile
(macrovascular complications)

Fasting lipid profile:

Serum cholesterol, serum triglycerides,

LDL cholesterol levels, HDL cholesterol levels

Tests for renal involvement (nephropathy)

Renal function: Measurement of albumin excretion in urine (microalbuminuria) – Earliest indicator of renal involvement

Serum creatinine is elevated in the later stages, when there is renal failure

Comparison of type 1 and type 2 diabetes mellitus

	Type 1 Diabetes	Type 2 Diabetes
AGE OF ONSET	Usually during childhood or puberty; symptoms develop rapidly	Frequently after age 35 years; symptoms develop gradually
NUTRITIONAL STATUS AT TIME OF DISEASE ONSET	Frequently undernourished	Obesity usually present
PREVALENCE	<10% of diagnosed diabetics	>90% of diagnosed diabetics
GENETIC PREDISPOSITION	Moderate	Very strong
DEFECT OR DEFICIENCY	β Cells are destroyed, eliminating production of insulin	Insulin resistance combined with inability of β cells to produce appropriate quantities of insulin
FREQUENCY OF KETOSIS	Common	Rare
PLASMA INSULIN	Low to absent	High early in disease; low to absent in disease of long duration
ACUTE COMPLICATIONS	Ketoacidosis	Hyperosmolar hyperglycemic state
RESPONSE TO ORAL HYPOGLYCEMIC DRUGS	Unresponsive	Responsive
TREATMENT	Insulin is always necessary	Diet, exercise, oral hypoglycemic drugs, insulin (may or may not be necessary); reduction of risk factors (weight reduction, smoking cessation, blood pressure control, treatment of dyslipidemia) is essential to therapy